Power Budget: Praan AQ Node

Moving-straight mode, controller board + example motor load

Result: the system draws about 14 W while moving straight. The controller needs 3.3 V ≥ 800 mA and 5 V ≥ 300 mA; motors set the main supply size.

Basis

- Moving straight is the highest-consumption mode because of the motors. The controller load is the same in every motion mode (Wi-Fi and sensors always on).
- Currents are typical datasheet values. Peaks are added as if they coincide (worst case), then multiplied by a factor of safety of 1.5.
- No motor was specified. The 12 V motor values are an example to show the method and must be replaced with real datasheet values.

Loads

Load	Rail	Avg (mA)	Peak (mA)	Basis
ESP32-S3 (Wi-Fi connected, modem sleep)	3.3 V	40.8	350	Datasheet; TX burst at max power every 2 s
microSD card	3.3 V	1.1	100	SD write current; 512 B write every 16 s
DS3231 RTC	3.3 V	0.13	0.65	Datasheet; temp. conversion every 64 s
SHT30	3.3 V	0.02	1.5	Datasheet; 16 ms measurement per second
BME280	3.3 V	< 0.01	0.71	Datasheet; 1 Hz forced mode
LED, I2C and SQW pull-ups	3.3 V	0.2	15.6	Ohm's law; LED duty capped at 25 %
PMS5003	5 V	100	150	Datasheet ≤ 100 mA; start-up peak assumed +50 %
2× drive motors (example)	12 V	1000	5000	Assumed: 0.5 A running, 2.5 A stall each

Rails and voltage outputs

Rail	Avg current	Avg power	Peak current	Peak × FOS 1.5	Required rating
3.3 V ±3 %	42 mA	0.14 W	468 mA	703 mA	≥ 800 mA
5 V ±5 %	100 mA	0.50 W	150 mA	225 mA	≥ 300 mA
12 V (example)	1.0 A	13.3 W*	5.0 A	7.5 A	≥ 7.5 A peak
Total, moving straight		**≈ 14.0 W**			

* Motor input power = 12 V × 1.0 A / 0.9 driver efficiency.

Energy usage

Case	Power	Energy
Controller only (sensors, Wi-Fi, logging), 24 h	0.64 W	15.3 Wh/day
Controller with PMS duty-cycled (60 s every 5 min)	0.24 W	5.8 Wh/day
Whole system, moving straight	≈ 14 W	≈ 14 Wh per hour

Takeaways

- **PMS5003 is 78 % of controller power.** Duty-cycling it cuts controller energy by 62 % and extends sensor life.
- **Motors dominate the system.** Battery and supply sizing depend on the motor datasheet, not the electronics.
- **Vendor extras:** 3.3 V must stay above 3.0 V during Wi-Fi TX bursts (brownout), and a power-good signal with ≥ 50 ms hold-up lets firmware flush SD on power loss. All figures to be confirmed on hardware.